Data quality problems of the four NetFlow datasets

What we fix before the merge, and what we fix after it

Ahmed Hamouda

with the ToN-IoT and BoT-IoT cross-check by Cyrine Chabbar

Phase 2 (data understanding) and phase 3 (cleaning)

How to read this document

This document lists every data quality problem found in the four NetFlow v3 datasets (NF-UNSW-NB15, NF-CSE-CIC-IDS2018, NF-ToN-IoT and NF-BoT-IoT), the evidence for it, and the decision taken. All four datasets were examined with the same checks, Steps 0–8 of phase 2 (notebooks `01_unsw`, `02_cic18`, `03_ton`, `04_bot`). The team approved every decision, and phase 3 applied them (notebooks `01_unsw_cleaning` to `04_bot_cleaning` and `06_merge_netflow`); where the cleaning measured a slightly different number, the entry says so.

Where each problem is listed. Every problem appears once, in the place where it is solved:

- **Sections 1–4, inside one dataset, before the merge.** Problems that remove or keep *rows*, problems that exist in only that dataset, and differences that would stop the four files from being stacked (column types, label names).
- **Section 5, on the merged table.** Problems shared by several datasets that one rule solves for all of them (column changes, value rules, which columns a model may use), and problems that only appear once the datasets are together.
- **Section 6, the plan.** How the cleaned data is reduced, balanced, split and evaluated, with its numbers.

How each entry is organised. Each section first gives the **decisions that change the data** (rows removed, values corrected, columns dropped or kept out of the model), then the **decisions to keep it as it is** (the problem is documented and nothing changes). Every entry shows the notebook **check** that found it, its **size**, a short explanation, a real example taken from the data, and the **decision** with its reason. **Row** in an example is the position of the flow in the original CSV file (the first data row is 0).

The order of phase 3. For each dataset: its own before-merge decisions → one row per different behaviour, with a `count` of the flows it stands for → `clean()` → `map_labels()` → save. Then the four files are stacked and the after-merge decisions are applied once.

Cross-check. NF-ToN-IoT-v3 and NF-BoT-IoT-v3 were analysed independently by Cyrine Chabbar and Ahmed Hamouda. Cyrine's notebooks are the main `03_ton` and `04_bot`, completed with the checks that only Ahmed's analysis had (marked *added from Ahmed Hamouda's analysis*). Every count measured in both analyses is identical. Cyrine's checks added four rules, now applied to all four datasets (K17, K18 and the exceptions in M7 and K7), and a documentation error (K16). Both analyses reached, on their own, the decision not to remove the ToN-IoT and BoT-IoT attack flows that look like Benign (T6, B7).

Dataset	Flows	Attack share	Change	Keep
NF-UNSW-NB15-v3	2,365,424	5.4%	4	4
NF-CSE-CIC-IDS2018-v3	20,115,529	12.9%	7	3
NF-ToN-IoT-v3	27,520,260	39.0%	5	4
NF-BoT-IoT-v3	16,933,808	99.7%	3	4
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1 NF-UNSW-NB15-v3 (2,365,424 flows) – before the merge

1.1 Decisions that change the data

U1. Strong class imbalance

Check: 1.4 class distribution | **Size:** Benign 94.6% (17.5 benign flows per attack flow)

Almost every flow is Benign. A model that always answers "Benign" would already be 94.6% accurate and still miss every attack.

Many flows are repeats of the same behaviour. Keeping each different behaviour once leaves 1,307,351 Benign and 97,749 attack rows, and no pattern is lost.

Example from the data

Attack	flows	different behaviours kept
Benign	2,237,731	1,307,351
Exploits	42,748	40,424
Fuzzers	33,816	26,987
Generic	19,651	5,726
Reconnaissance	17,074	11,908
DoS	5,980	5,360
Backdoor	4,659	3,656
Shellcode	2,381	2,316
Analysis	1,226	1,226
Worms	158	146
total	2,365,424	1,405,100

Decision: Delete nothing by hand. Inside the dataset, exact repeats of a behaviour are collapsed into one row with a `count` column, so every different behaviour is kept (section 6). Training balances classes and networks with weights; models are judged with macro-F1, per-class recall and a realistic 95% / 5% test set, not accuracy.

Why before the merge: The table of different behaviours (one row each, with its repeat count) is built inside each dataset, after its own removals and before the four files are stacked (section 6).

Other datasets: CIC-2018 (Benign 87.1%), ToN-IoT (61.0%); BoT-IoT is the opposite (Benign 0.3%).

U2. Missing and infinite values in the `..._SECOND_BYTES` columns

Check: 3.1–3.4 NaN / inf, cause and class | **Size:** 63,425 NaN + 181,561 inf; 75.8% of Generic

Both columns are a byte count divided by the flow duration. When the duration is 0 ms (122,493 flows), the division gives NaN (0 ÷ 0) or infinity (bytes ÷ 0).

Check 3.3 also showed the columns are mislabelled: the values are bytes per **millisecond**, not per second, and the two directions are swapped.

The missing values are not random: 75.8% of Generic flows are affected, but only 3.9% of Benign.

Nothing is lost by dropping them: `SRC_TO_DST_AVG_THROUGHPUT` (IN bytes × 8 ÷ duration) and `DST_TO_SRC_AVG_THROUGHPUT` (OUT bytes × 8 ÷ duration) measure the same speed per direction, correctly named, in bits per second, and never NaN or infinite (the exporter divides by the duration + 1 ms). Each `SECOND_BYTES` column matches the throughput of the **opposite** name (correlation 0.996 in UNSW), about 8,000 times smaller (8 bits × 1,000 ms), as the second table shows for row 588977 (701 bytes in, 1,052 bytes out, 63 ms).

Example from the data

Row	IN_BYTES	OUT_BYTES	FLOW_DURATION_MILLISECONDS	SRC_TO_DST_SECOND_BYTES	DST_TO_SRC_SECOND_BYTES	Attack
40	1,064	0	0	NaN	inf	Fuzzers
63	130	162	0	inf	inf	Benign

Attack	% of the class with NaN or inf
Generic	75.8
Reconnaissance	33.9
Shellcode	33.4
Backdoor	25.9

Row	IN_BYTES	OUT_BYTES	FLOW_DURATION_MILLISECONDS	SRC_TO_DST_SECOND_BYTES	DST_TO_SRC_AVG_THROUGHPUT	DST_TO_SRC_SECOND_BYTES	SRC_TO_DST_AVG_THROUGHPUT
588977	701	1,052	63	16.698	131,500	11.127	87,625

Decision: **Drop both columns** before `clean()` runs. No imputation and no row deletion.

Why before the merge: `clean()` deletes every row that still has a NaN, and it runs on each dataset before the merge. The columns must therefore be dropped first, in this dataset. Otherwise 75.8% of the Generic class would silently disappear.

Other datasets: BoT-IoT has the same NaN/inf (it would lose 65% of its Benign); CIC-2018 and ToN-IoT store the columns as integers without NaN, but with the same wrong formula.

U3. `L7_PROTO` stores two codes in one decimal number

Check: 2.4 `L7_PROTO` decimals + raw CSV text | **Size:** 23,866 flows with a sub-code, 8 ambiguous

`L7_PROTO` holds nDPI protocol identifiers. The numbers are documented in nDPI's `ndpi_protocol_ids.h` (github.com/ntop/nDPI): 7 = HTTP, 37 = BitTorrent, 10 = NetBIOS, 16 = SMBv1, 253 = SOAP, 376 = WebDAV. nDPI describes a flow by a *master* protocol (the carrier) and an *application* protocol (what runs inside it), written with a dot in protocol names such as `TLS.YouTube`.

We therefore read `7.37` as "BitTorrent carried over HTTP". The numeric dot format is not described in an official document, so this reading is an interpretation, but the data supports it: the `7.37` flow below goes to port 80 (HTTP), `10.16` to port 445 (SMB), and SOAP (253) is a protocol that runs inside HTTP.

Because the column is a float, trailing zeros were lost when the dataset was written: `7.5` can be HTTP+DNS (5) or HTTP+RTSP (50). This affects 8 flows (all `7.5`; the phase 2 notebook estimated 10, the cleaning counted exactly). CIC-2018 and ToN-IoT store the same column as an integer.

Example from the data

Row	PROTOCOL	L4_DST_PORT	L7_PROTO	Attack	raw text in the CSV
302	6	445	10.16	Exploits	10.16
1036	6	80	7.37	Benign	7.37
4035	6	7001	7.253	Exploits	7.253
34960	6	80	7.5	Generic	7.5

Decision: Split into two integer columns before the merge: `L7_MAIN` (before the dot, always reliable) and `L7_SUB` (after the dot, 0 if none). Document the 8 ambiguous flows.

Why before the merge: A merged column must have one type and one meaning. The fix needs this dataset's own values, so it is done before the files are stacked. The split needs the raw text of this file.

Other datasets: BoT-IoT has the same decimals (1,851 flows, including 796 ambiguous `7.5`); CIC-2018 and ToN-IoT only need `L7_SUB = 0`.

U4. Exact duplicate rows

Check: 5.1 exact duplicates | **Size:** 14,815 rows (0.63%), 14,801 of them Benign

These rows are identical in all 55 columns, including the IP addresses and the start time in milliseconds. They are the same flow written twice. They add no information and could end up in both the training and the test set, which makes the model look better than it is.

Example from the data

Row	FLOW_START_MILLISECONDS	IPV4_SRC_ADDR	L4_SRC_PORT	L4_DST_PORT	IN_BYTES	Attack
2305	1424239673765	149.171.126.18	1043	53	132	Benign
2306	1424239673765	149.171.126.18	1043	53	132	Benign

Decision: Remove them, keeping the first copy (shared `clean()` rule).

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: CIC-2018 (628,474, whole attack captures recorded twice), ToN-IoT (1,816,137); none in BoT-IoT.

1.2 Decisions to keep it as it is

U5. Very rare class: Worms

Check: 1.4 class distribution | **Size:** 158 flows (0.007%)

Worms is real attack traffic but too small to learn reliably at label level. The two next smallest labels are also small.

Example from the data

Attack	flows
Worms	158
Analysis	1,226
Shellcode	2,381
Backdoor	4,659

Decision: Keep every flow; give the class a higher training weight. Report results also at family level: Worms + Backdoor = *Malware and botnets*, 4,817 flows.

Why before the merge: Rare labels are a property of this dataset; removing them would lose the only Worms data of the project.

Other datasets: Theft in BoT-IoT (1,615), SQL injection / XSS in CIC-2018 (440 / 480), MITM / Ransomware in ToN-IoT.

U6. Only 6 of the 9 attack families are present

Check: 1.3 taxonomy mapping | **Size:** 4 families missing

UNSW has no Credential attacks, Web and injection, Interception and spoofing or Data theft. It is, however, the only source of Fuzzing and crypto (Fuzzers, Generic) and of Exploits and Shellcode.

Example from the data

Family	flows
Benign	2,237,731
Fuzzing and crypto	53,467
Exploitation and intrusion	45,129
Reconnaissance	18,300
Denial of service	5,980
Malware and botnets	4,817

Decision: No change to the data. Documented: the missing families come from the other datasets.

Why before the merge: It describes what this dataset can teach.

Other datasets: Each NetFlow dataset covers a different set of families (see M12).

U7. Invalid codes, all in attack traffic

Check: 2.5 invalid codes | **Size:** 754 flows with unassigned protocol numbers + 6 port 0 + 6 invalid DNS + 6 invalid ICMP types

Protocol numbers 148–252 are not assigned by IANA, port 0 is reserved, and some DNS/ICMP type values do not exist. Normally these would be recording errors.

Here **no Benign flow** uses them: 620 of the 754 unassigned-protocol flows are Fuzzers, which send random, malformed packets on purpose. The invalid value *is* the attack.

Example from the data

Row	PROTOCOL	IN_PKTS	IN_BYTES	Attack
104904	246	2	200	Fuzzers
104922	247	2	200	Fuzzers
104943	248	2	200	Fuzzers

Decision: Keep them unchanged.

Why before the merge: Only UNSW has them, and they belong to its attack classes.

Other datasets: ToN-IoT and BoT-IoT have a few invalid codes in Benign traffic (see K7).

U8. Same behaviour, different attack label

Check: 5.4 behaviour duplicates | **Size:** 30,907 rows in 993 groups (71% of Generic); never Benign vs attack

Leaving out the identifiers (IPs, source port, times), these flows are identical in every column, yet carry different attack labels. In the example, the same 2-packet DNS request with no reply appears 12,860 times as Generic and 230 times as Backdoor.

The label describes the whole attack campaign, while one flow shows only one small step of it: many attacks start with the same probe. No model can separate these flows, so label-level accuracy has a ceiling.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
2226	17	53	2	114	0	0	0	0	0	Generic
156587	17	53	2	114	0	0	0	0	0	Backdoor

Decision: Keep them. Report results at label level and at family level; the binary task (attack or not) is not affected, because every flow here is an attack.

Why before the merge: The conflict is between labels of this dataset only.

Other datasets: BoT-IoT has the same between DoS and DDoS (B6); ToN-IoT and CIC-2018 also have conflicts with Benign (T6, C7).

2 NF-CSE-CIC-IDS2018-v3 (20,115,529 flows) – before the merge

2.1 Decisions that change the data

C1. Labels missing from the taxonomy, with inconsistent spellings

Check: 1.1 spellings + 1.3 taxonomy | **Size:** 11 of 15 labels, 2,208,048 flows

`map_labels` stops on 11 labels because they are not in the team taxonomy. Their spelling is also inconsistent: `DDOS_attack-` next to `DDoS_attacks-`, `-Bruteforce` next to `-BruteForce`, the typo `Infilteration`, and `Brute_Force_-XSS` for an XSS attack.

The tool name (HOIC, Hulk, Slowloris, ...) is useful detail, so it is kept in the clean label.

Example from the data

raw label	clean label	family	flows
DDOS_attack-HOIC	DDoS HOIC	Denial of service	1,032,311
DDoS_attacks-LOIC-HTTP	DDoS LOIC-HTTP	Denial of service	288,589
SSH-Bruteforce	SSH brute force	Credential attacks	188,474
Brute_Force_-XSS	XSS	Web and injection	480
Infilteration	Infiltration	Exploitation and intrusion	188,152

Decision: Add the mapping to `config.py` (after team agreement). `attack_label` gets the clean name, `attack_family` the family; the raw `Attack` column is not edited.

Why before the merge: The four files can only be stacked once every label has one clean name and one family; the spelling of a label is a property of this dataset.

Other datasets: ToN-IoT writes its labels in lower case (T1); UNSW and BoT-IoT were already clean.

C2. Class imbalance and very large attack classes

Check: 1.4 class distribution | **Size:** Benign 87.1%; DDoS HOIC 1,032,311

Benign dominates, and some attack tools produced over a million flows while SQL injection has 440.

After this dataset's removals (C5–C7), keeping each different behaviour once leaves 8,178,232 Benign and 767,560 attack rows. FTP-BruteForce shows why repeats add nothing: 386,720 flows, but one single behaviour (C10).

Example from the data

Attack	flows	different behaviours kept
Benign	17,514,626	8,178,232
DDoS HOIC	1,032,311	244,060
FTP brute force	386,720	1
DDoS LOIC-HTTP	288,589	265,826
Bot	207,703	2,804
SSH brute force	188,474	93,979
Infiltration	188,152	27,213
DoS SlowHTTPTest	105,550	0
DoS Hulk	100,076	100,071
DoS GoldenEye	61,300	27,410
DoS Slowloris	36,040	3,510
DDoS LOIC-UDP	3,450	1,723
Web brute force	1,618	910
XSS	480	0
SQL injection	440	53
total	20,115,529	8,945,792

Decision: Delete nothing by hand. Inside the dataset, exact repeats of a behaviour are collapsed into one row with a `count` column, so every different behaviour is kept (section 6). Training balances classes and networks with weights; models are judged with macro-F1, per-class recall and a realistic 95% / 5% test set, not accuracy.

Why before the merge: The table of different behaviours (one row each, with its repeat count) is built inside each dataset, after its own removals and before the four files are stacked (section 6).

Other datasets: Same in UNSW (U1) and ToN-IoT (T2); inverted in BoT-IoT (B1).

C3. Three columns have a different type than in UNSW and BoT-IoT

Check: 0.4 column comparison + 3.3 formula | **Size:** `L7_PROTO`, `SRC_TO_DST_SECOND_BYTES`, `DST_TO_SRC_SECOND_BYTES`

The four files have the same 55 columns in the same order, but these three are integers here and decimals in UNSW and BoT-IoT. Stacked as they are, pandas would turn them into floats and mix two representations.

The `..._SECOND_BYTES` columns also carry the wrong meaning: `SRC_TO_DST_SECOND_BYTES` is reproduced by **OUT**_BYTES ÷ duration (direction swapped, per millisecond) for 79% of the flows, as the example shows.

Example from the data

Column	UNSW	CIC-2018	ToN-IoT	BoT-IoT
L7_PROTO	decimal	integer	integer	decimal
SRC_TO_DST_SECOND_BYTES	decimal	integer	integer	decimal
DST_TO_SRC_SECOND_BYTES	decimal	integer	integer	decimal

Row	L7_PROTO	IN_BYTES	OUT_BYTES	FLOW_DURATION_MILLISECONDS	SRC_TO_DST_SECOND_BYTES	DST_TO_SRC_SECOND_BYTES	floor(OUT_BYTES ÷ duration)
4	7	373	700	1	700	373	700
9	5	73	140	63	2	1	2

Decision: Drop the two `..._SECOND_BYTES` columns (as in UNSW U2). Write `L7_PROTO` as `L7_MAIN` = the value and `L7_SUB` = 0, the same representation as UNSW and BoT-IoT.

Why before the merge: A merged column must have one type and one meaning. The fix needs this dataset's own values, so it is done before the files are stacked.

Other datasets: Same in ToN-IoT (T3).

C4. Exact duplicate rows: whole attack captures recorded twice

Check: 5.1 exact duplicates | **Size:** 628,474 rows (3.12%)

Every flow of FTP-BruteForce, SSH-Bruteforce, DoS GoldenEye, DoS Slowloris and DDoS LOIC-UDP appears **exactly twice**, identical in all 55 columns, including the millisecond timestamp. Those capture files were written into the dataset twice.

Kept, these copies would double the weight of these attacks and leak between training and test sets.

Example from the data

Row	FLOW_START_MILLISECONDS	IPV4_SRC_ADDR	L4_SRC_PORT	L4_DST_PORT	Attack
396828	1518618806718	18.221.219.4	39922	21	FTP-BruteForce
396829	1518618806718	18.221.219.4	39922	21	FTP-BruteForce

Attack	duplicate rows
FTP brute force	193,360
Benign	121,872
Bot	96,243
SSH brute force	94,237
Infiltration	72,347

Decision: Remove them, keeping the first copy (shared `clean()` rule).

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: UNSW (14,815), ToN-IoT (1,816,137); none in BoT-IoT.

C5. The same flow carries two different labels

Check: 5.2 identical flows with a different label | **Size:** 880 rows

These rows are identical in every column, including the IP addresses and the millisecond start time, so they are the same flow, yet one copy is labelled XSS and the other SQL injection or web brute force. The three web attacks were run from the same machine against the same server at the same time.

Example from the data

Row	FLOW_START_MILLISECONDS	IPV4_SRC_ADDR	IPV4_DST_ADDR	L4_DST_PORT	IN_BYTES	Attack
10150628	1519302310148	52.14.77.172	172.31.69.28	500	528	Web brute force
20115069	1519302310148	52.14.77.172	172.31.69.28	500	528	XSS

Decision: Remove **all** copies of these flows: the correct label cannot be known.

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: Not found in UNSW, ToN-IoT or BoT-IoT.

C6. DoS SlowHTTPTest is mislabelled: its flows are refused FTP connections

Check: 5.4 behaviour + profile | **Size:** 105,550 flows (+ 60 port-21 flows labelled SSH-Bruteforce)

SlowHTTPTest is an HTTP attack: it keeps web connections open with very slow requests. But **every** one of its flows goes to port 21 (FTP), sends 1 packet with SYN (client flags 2) and receives a reset (server flags 20). There is no HTTP traffic at all.

The flows are identical to those of FTP-BruteForce, which was captured from the same attacker. A model would learn a wrong picture of a DoS attack.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	OUT_PKTS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
4143941	6	21	1	1	2	20	DoS_attacks-SlowHTTPTest
396828	6	21	1	1	2	20	FTP-BruteForce

Decision: Remove these flows.

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: Not found in the other datasets; ToN-IoT has a similar case (T5).

C7. Attack labels on normal traffic

Check: 5.4 attack flows identical to Benign + same-machine check | **Size:** 150,600 attack flows = 5.8% of CIC-2018's attacks (Infiltration 75%, XSS 91%, SQL injection 78%, web brute force 41%)

These attack flows are identical in behaviour to Benign flows of the same network: in the first example a 1-packet DNS lookup is labelled Benign in one row and Infiltration in the other.

The strongest evidence is the same machine. For 55% of the removed Infiltration flows, the same victim computer sent the identical flow labelled Benign at another time; for 45% even on a day without any infiltration attack, and for 32% on the same day. The second table follows computer 172.31.69.12: its DNS lookup to 172.31.0.2 (the DNS server AWS gives every machine in the network) is Benign on every capture day, and becomes "Infiltration" only on 28 February and 1 March, the two infiltration days.

So the label followed the attack's time window, not the traffic: CIC-2018 labelled everything an infected machine did during the attack, including its normal background traffic. The flows that are kept look different (third table): fewer DNS lookups, more packets, and ports used to move between Windows computers (SMB 445, RPC 135).

NetFlow has no packet content, so a few removed flows (e.g. a DNS lookup made by the malware) could have been a tiny real part of the attack; they are identical to normal flows, so no model could separate them anyway.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
66	17	53	1	64	1	80	1	0	0	Benign
14116240	17	53	1	64	1	80	1	0	0	Infilteration

Day	Benign	Infiltration
2018-02-14	6	0
2018-02-15	2	0
2018-02-16	4	0
2018-02-20	6	0
2018-02-21	14	0
2018-02-22	4	0
2018-02-28	2	8
2018-03-01	2	8
2018-03-02	2	0

(continued)

	removed (identical to Benign)	**kept**
Infiltration flows	140,826	47,326
to the DNS port 53	64%	29%
median packets in	1	5
median bytes in	65	462

Attack	**flows**	**identical to Benign**	**left if removed**
Infiltration	188,152	140,826	47,326
DDoS LOIC-HTTP	288,589	8,325	280,264
Web brute force	1,618	668	950
XSS	480	436	44
SQL injection	440	344	96
DoS Hulk	100,076	1	100,075

Decision: Remove the attack flows whose behaviour also appears as Benign in CIC-2018 (relabelling them as Benign would also be defensible, but would invent labels; CIC-2018 already has 17.5 million Benign flows). C7 alone leaves XSS with 44 flows and SQL injection with 96; together with C5 (whose 880 conflicting rows are exactly these web attacks), **no XSS flow remains** and SQL injection keeps 53 different behaviours. CIC-2018 therefore contributes almost nothing to *Web and injection*; ToN-IoT provides that family.

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: ToN-IoT and BoT-IoT have the same pattern but for a different reason, and keep their flows (T6, B7).

2.2 Decisions to keep it as it is

C8. Very small classes

Check: 1.4 class distribution | **Size:** SQL injection 440, XSS 480, web brute force 1,618, DDoS LOIC-UDP 3,450

Four labels have too few flows to learn reliably at label level, and C7 makes the web attacks even smaller.

Example from the data

Attack	flows
SQL injection	440
XSS	480
Web brute force	1,618
DDoS LOIC-UDP	3,450

Decision: Keep them all; report results also at family level.

Why before the merge: Rare labels are a property of this dataset.

Other datasets: UNSW Worms (U5), ToN-IoT MITM / Ransomware (T8), BoT-IoT Theft (B4).

C9. Only 6 of the 9 attack families are present

Check: 1.3 family totals | **Size:** missing: Reconnaissance, Fuzzing, Interception, Data theft

CIC-2018 adds *Credential attacks* and *Web and injection*, which UNSW lacks, and is the only source of Bot and Infiltration.

Example from the data

Family	flows
Benign	17,514,626
Denial of service	1,627,316
Credential attacks	576,812
Malware and botnets	207,703
Exploitation and intrusion	188,152
Web and injection	920

Decision: No change; documented.

Why before the merge: It describes what this dataset can teach.

Other datasets: See M12 for the family coverage of the merged data.

C10. FTP-BruteForce contains only refused connections

Check: 5.4 FTP profile | **Size:** 386,720 flows, 1 distinct behaviour

A brute-force attack should try many passwords on a login. Here every flow is the same single SYN packet to port 21 answered by a reset: the FTP server was closed, so no password was ever tried. After removing the duplicates (C4) 193,360 identical flows remain.

Example from the data

Row	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
396828	21	1	60	1	2	20	FTP-BruteForce
396829	21	1	60	1	2	20	FTP-BruteForce

Decision: Keep them and document that they show failed connection attempts, not password guessing.

Why before the merge: It is still attack traffic (a scan of a closed port), but only one pattern.

Other datasets: Not found in the other datasets.

3 NF-ToN-IoT-v3 (27,520,260 flows) – before the merge

3.1 Decisions that change the data

T1. Inconsistent label spelling

Check: 1.1 label spellings | **Size:** 10 labels

Eight labels are written in lower case (`ddos`, `xss`, `password`, ...) while `Benign` and `Backdoor` are capitalised. In the merged data `dos` (ToN-IoT) and `DoS` (UNSW, BoT-IoT) would be two different classes. The taxonomy lookup ignores case, so every label maps.

Example from the data

raw label	clean label	family	flows
Benign	Benign	Benign	16,792,214
ddos	DDoS	Denial of service	4,141,256
xss	XSS	Web and injection	2,834,435
password	Password	Credential attacks	1,594,777
scanning	Scanning	Reconnaissance	1,358,977
injection	Injection	Web and injection	381,777
dos	DoS	Denial of service	203,456
Backdoor	Backdoor	Malware and botnets	203,384
mitm	MITM	Interception and spoofing	6,013
ransomware	Ransomware	Malware and botnets	3,971

Decision: Map every label to its clean name and family with `map_labels` before the merge; the raw `Attack` column is not edited.

Why before the merge: The four files can only be stacked once every label has one clean name and one family; the spelling of a label is a property of this dataset.

Other datasets: CIC-2018 has the same kind of problem (C1).

T2. Class imbalance and very large attack classes

Check: 1.4 class distribution | **Size:** Benign 61.0%; DDoS 4,141,256; XSS 2,834,435

ToN-IoT is the largest file (27.5 million flows). Benign dominates, but several attack labels have millions of flows, while MITM has 6,013 and Ransomware 3,971.

ToN-IoT's attacks are very varied: after T5, keeping each different behaviour once still leaves 3,355,000 attack rows, next to 2,282,642 Benign rows.

Example from the data

Attack	flows	different behaviours kept
Benign	16,792,214	2,282,642
DDoS	4,141,256	451,474
XSS	2,834,435	1,497,451
Password	1,594,777	697,124
Scanning	1,358,977	225,074
Injection	381,777	333,652
DoS	203,456	121,803
Backdoor	203,384	20,641
MITM	6,013	4,951
Ransomware	3,971	2,830
total	27,520,260	5,637,642

Decision: Delete nothing by hand. Inside the dataset, exact repeats of a behaviour are collapsed into one row with a `count` column, so every different behaviour is kept (section 6). Training balances classes and networks with weights; models are judged with macro-F1, per-class recall and a realistic 95% / 5% test set, not accuracy.

Why before the merge: The table of different behaviours (one row each, with its repeat count) is built inside each dataset, after its own removals and before the four files are stacked (section 6).

Other datasets: UNSW (U1), CIC-2018 (C2); inverted in BoT-IoT (B1).

T3. Three columns have a different type than in UNSW and BoT-IoT

Check: 0.4 column comparison + 3.3 formula | **Size:** `L7_PROTO`, `SRC_TO_DST_SECOND_BYTES`, `DST_TO_SRC_SECOND_BYTES`

As in CIC-2018, these columns are integers here and decimals in UNSW and BoT-IoT. `L7_PROTO` never has a sub-code.

`SRC_TO_DST_SECOND_BYTES` is reproduced by floor(**OUT**_BYTES ÷ duration in ms) for 74.6% of the flows: the same swapped direction and wrong unit; 72% of its values are 0.

Example from the data

Row	L7_PROTO	IN_BYTES	OUT_BYTES	FLOW_DURATION_MILLISECONDS	SRC_TO_DST_SECOND_BYTES	floor(OUT_BYTES ÷ duration)
1546	4	866	1,658	56	29	29
1572	4	866	1,658	65	25	25

Decision: Same as CIC-2018 C3: drop the two `..._SECOND_BYTES` columns and write `L7_PROTO` as `L7_MAIN` + `L7_SUB = 0`.

Why before the merge: A merged column must have one type and one meaning. The fix needs this dataset's own values, so it is done before the files are stacked.

Other datasets: CIC-2018 (C3); UNSW and BoT-IoT have the decimal version (U2, U3, B2, B3).

T4. Exact duplicate rows, some repeated hundreds of times

Check: 5.1 exact duplicates | **Size:** 1,816,137 rows (6.60%)

Rows identical in all 55 columns. Most are Benign: one flow appears **437 times** (the example shows three of its copies, same millisecond, same ports). Every Backdoor flow appears exactly twice (101,692 copies), and 745 MITM flows are repeated.

Repeated rows give one flow hundreds of times the weight of the others and leak between training and test sets.

Example from the data

Row	FLOW_START_MILLISECONDS	IPV4_SRC_ADDR	L4_SRC_PORT	IPV4_DST_ADDR	L4_DST_PORT	Attack
25886380	1556542513555	192.168.1.1	53	192.168.1.169	52970	Benign
25886381	1556542513555	192.168.1.1	53	192.168.1.169	52970	Benign
25886382	1556542513555	192.168.1.1	53	192.168.1.169	52970	Benign

Attack	duplicate rows
Benign	1,713,700
Backdoor	101,692
MITM	745

Decision: Remove them, keeping the first copy (shared `clean()` rule).

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: UNSW (14,815), CIC-2018 (628,474); none in BoT-IoT.

T5. Web attacks labelled on DNS traffic

Check: 5.4 label profile (main destination ports) + protocol check | **Size:** 1,261,353 flows (XSS 1,244,884 = 44% of XSS, Injection 16,469)

XSS and injection are web attacks: the malicious script or command travels inside an HTTP request to a web server, and HTTP runs over **TCP** on port 80/443. Yet 44% of the XSS flows go to port 53, the DNS port.

The protocol check settles it (second table). The XSS flows to web ports look like a web attack: TCP, several packets, about 1.8 KB. The XSS flows to port 53 are **UDP**, recognised by nDPI as **DNS**, one packet of about 68 bytes, the same shape as Benign DNS lookups. A web attack cannot be carried by a single UDP DNS query.

(The number 53 can of course appear *inside* an XSS payload, e.g. `alert(53)`, but NetFlow never sees the payload: here 53 is the destination port of the connection.) As in CIC-2018 (C7), the attacker machine's own background DNS lookups were labelled with the attack. A model would learn "DNS query = XSS".

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack	L7_PROTO
21963056	17	53	1	65	1	65	5	0	0	xss	5
21963057	17	53	1	65	1	81	3	0	0	xss	5
2048192	17	53	1	65	1	1,196	12	0	0	Benign	5

	flows	TCP %	UDP %	recognised as DNS (L7 5) %	median packets in	median bytes in
XSS to web ports 80/443	1,589,533	100	0	0	6	1,772
XSS to port 53	1,244,884	0	100	100	1	65
Injection to port 53	16,469	0	100	100	1	65
Benign to port 53	913,131	0.7	99.3	100	1	71

Decision: Remove the XSS and injection flows to port 53. Relabelling them as Benign would invent a label; ToN-IoT already has 16.8 million Benign flows, many of them DNS. XSS keeps 1,589,551 flows and injection 365,308, both far above what is needed.

Why before the merge: It removes or keeps rows of this dataset only, and has to happen before the table of different behaviours is built (section 6); otherwise wrong rows would be counted as real behaviours.

Other datasets: Same cause as CIC-2018 C6/C7. DoS and DDoS flows to port 53 are kept: a DNS flood is a real DoS attack. Not found in Cyrine Chabbar's independent ToN-IoT check (her check did not profile ports per label).

3.2 Decisions to keep it as it is

T6. Many attack flows look exactly like Benign flows

Check: 5.4 attack flows identical to Benign | **Size:** 6.1 million attack flows; Scanning 94.3%, DDoS 76.3%

Most ToN-IoT flows are tiny: the median flow has 1 packet in each direction. A port scan sends one SYN and gets no answer; a normal client that reaches a closed port does exactly the same. In the example, the same 1-packet, 44-byte SYN to port 443 is Benign in one row and Scanning in the other.

Here the label is not wrong, the flow is simply too short to show the difference: a scan is visible as **many** such flows, not in one of them. Applying the CIC-2018 rule (C7) would delete 94% of Scanning and 76% of DDoS.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
275601	6	443	1	44	0	0	1	2	0	Benign
2316	6	443	1	44	0	0	1	2	0	scanning

Attack	flows	identical to Benign	%
Scanning	1,358,977	1,281,266	94.3
DDoS	4,141,256	3,160,296	76.3
XSS	2,834,435	1,046,970	36.9
Password	1,594,777	531,206	33.3
DoS	203,456	67,354	33.1
Ransomware	3,971	355	8.9
MITM	6,013	508	8.4
Injection	381,777	28,752	7.5
Backdoor	203,384	2,530	1.2

Decision: Keep them; **do not** apply the CIC-2018 rule here. Document it as the limit of single-flow features; results on these classes are reported with this caveat.

Why before the merge: The decision depends on why the conflict exists in this dataset (wrong label in CIC-2018, too-short flow here).

Other datasets: CIC-2018 removes them because the labels are wrong (C7); BoT-IoT keeps them for the same reason as here (B7). Cyrine Chabbar's independent ToN-IoT check reached the same decision with the same counts ("removal is not justified by behavioural overlap alone").

T7. Same behaviour, different attack label

Check: 5.4 behaviour duplicates | **Size:** e.g. 1,671,924 DDoS and 695 Injection flows share one behaviour

A 2-packet, 112-byte connection to port 443 answered by one packet is labelled DDoS, Injection or XSS depending on the row. Each attack opens connections like this; the difference lies in the payload, which NetFlow does not record.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
10764488	6	443	2	112	1	60	1	18	18	ddos
10425763	6	443	2	112	1	60	1	18	18	injection

Decision: Keep them; report label-level and family-level results separately.

Why before the merge: The conflict is between labels of this dataset only.

Other datasets: UNSW (U8), BoT-IoT DoS vs DDoS (B6).

T8. Rare classes

Check: 1.4 class distribution | **Size:** MITM 6,013, Ransomware 3,971

ToN-IoT is the only source of MITM and Ransomware, and both are small.

Example from the data

Attack	flows
MITM	6,013
Ransomware	3,971

Decision: Keep every flow; report at family level too.

Why before the merge: Rare labels are a property of this dataset.

Other datasets: UNSW Worms (U5), CIC-2018 web attacks (C8), BoT-IoT Theft (B4).

T9. Only 6 of the 9 attack families are present

Check: 1.3 family totals | **Size:** missing: Exploitation, Fuzzing, Data theft

ToN-IoT is the widest IoT source: it is the only one with *Interception and spoofing* (MITM), and the largest source of *Web and injection* and *Credential attacks*.

Example from the data

Family	flows
Benign	16,792,214
Denial of service	4,344,712
Web and injection	3,216,212
Credential attacks	1,594,777
Reconnaissance	1,358,977
Malware and botnets	207,355
Interception and spoofing	6,013

Decision: No change; documented.

Why before the merge: It describes what this dataset can teach.

Other datasets: See M12.

4 NF-BoT-IoT-v3 (16,933,808 flows) – before the merge

4.1 Decisions that change the data

B1. Inverted imbalance: almost no Benign traffic

Check: 1.4 class distribution | **Size:** Benign 51,989 flows (0.3%)

BoT-IoT is 99.7% attacks: two flood labels alone have 15 million flows. Here Benign is the rare class, and its flows are unusual (65% have a zero duration, see B2).

The floods are extremely repetitive: the 16.9 million attack flows contain only 1,541,364 different behaviours (91% are repeats), so keeping each behaviour once loses no attack pattern. Benign keeps 21,951 rows.

Example from the data

Attack	flows	different behaviours kept
DoS	8,034,190	354,734
DDoS	7,150,882	287,453
Reconnaissance	1,695,132	898,729
Benign	51,989	21,951
Theft	1,615	448
total	16,933,808	1,563,315

Decision: Delete nothing by hand. Inside the dataset, exact repeats of a behaviour are collapsed into one row with a `count` column, so every different behaviour is kept (section 6). Training balances classes and networks with weights; models are judged with macro-F1, per-class recall and a realistic 95% / 5% test set, not accuracy. Use BoT-IoT mainly as attack data and report Benign recall separately; do not use it to describe normal traffic.

Why before the merge: The table of different behaviours (one row each, with its repeat count) is built inside each dataset, after its own removals and before the four files are stacked (section 6).

Other datasets: The opposite of UNSW (U1), CIC-2018 (C2) and ToN-IoT (T2).

B2. Missing and infinite values in the `..._SECOND_BYTES` columns

Check: 3.1–3.4 NaN / inf, cause and class | **Size:** 387,350 NaN + 1,716,860 inf, all from 1,052,105 zero-duration flows

Exactly as in UNSW: a duration of 0 ms gives $0 \div 0 =$ NaN or bytes $\div 0 =$ infinity. The formula check confirms the swap: `SRC_TO_DST_SECOND_BYTES` = OUT_BYTES ÷ duration for 100% of the other flows.

The missing values hit the small classes: 64.9% of Benign and 70.3% of Theft. Because `clean()` drops every row with a NaN, it would delete most of BoT-IoT's normal traffic.

Example from the data

Row	IN_BYTES	OUT_BYTES	FLOW_DURATION_MILLISECONDS	SRC_TO_DST_SECOND_BYTES	DST_TO_SRC_SECOND_BYTES	Attack
72	229	0	0	NaN	inf	Benign
2044	71	126	0	inf	inf	Benign

Attack	% of the class with NaN or inf
Theft	70.3
Benign	64.9
Reconnaissance	41.2
DoS	2.8
DDoS	1.3

Decision: **Drop both columns** before `clean()` runs, as in UNSW.

Why before the merge: `clean()` deletes every row that still has a NaN, and it runs on each dataset before the merge. The columns must therefore be dropped first, in this dataset. Otherwise 65% of Benign and 70% of Theft would disappear.

Other datasets: UNSW (U2); CIC-2018 and ToN-IoT store them as integers (C3, T3). Cyrine Chabbar's BoT-IoT check found the same 387,350 missing cells and proposed to avoid blanket row deletion; the infinite values and the swapped formula complete the picture and lead to dropping the columns.

B3. `L7_PROTO` stores two codes in one decimal number

Check: 2.4 `L7_PROTO` decimals | **Size:** 1,851 flows, 796 of them ambiguous (`7.5`)

As in UNSW, `L7_PROTO` is a float: `10.16` is NetBIOS carrying SMB, `5.239` is DNS carrying another protocol. `7.5` (796 flows) can be HTTP+DNS (5) or HTTP+RTSP (50), because the trailing zero was lost. The cleaning found one more problem: BoT-IoT writes every value as a decimal (`7.0` = HTTP, no sub-code), and 30 values carry floating-point noise from the export (`5.2120000000000015` is 5.212). nDPI codes have at most three digits, so a longer decimal part is rounded to three.

Example from the data

Row	PROTOCOL	L4_DST_PORT	L7_PROTO	Attack
68	17	53	5.239	Benign
72	17	138	10.16	Benign
2044	17	53	5.212	Benign
17819	6	443	91.239	Benign

Decision: Split into `L7_MAIN` and `L7_SUB`, as in UNSW; document the ambiguous `7.5`.

Why before the merge: A merged column must have one type and one meaning. The fix needs this dataset's own values, so it is done before the files are stacked.

Other datasets: UNSW (U3).

4.2 Decisions to keep it as it is

B4. Very rare class: Theft, and wrong official numbers

Check: 0.1 row count + 1.4 class distribution | **Size:** Theft 1,615 flows (0.01%)

Theft (data exfiltration, keylogging) is the only *Data theft* label of the project, and it is tiny.

The official dataset page gives 16,993,808 flows and 1,651 Theft flows; the file contains 16,933,808 and 1,615. The checksum of the file matches the official manifest, so the page has typos (two swapped digits).

Example from the data

Count	official page	file
all flows	16,993,808	16,933,808
Theft	1,651	1,615

Decision: Keep every Theft flow; use the counts from the file and report the typo.

Why before the merge: It is a property of this dataset.

Other datasets: UNSW Worms (U5), ToN-IoT MITM / Ransomware (T8).

B5. Only 3 of the 9 attack families are present

Check: 1.3 family totals | **Size:** Denial of service, Reconnaissance, Data theft

BoT-IoT is almost entirely floods (DoS and DDoS, 15.2 million flows) and scans. It is the only source of Data theft.

Example from the data

Family	flows
Denial of service	15,185,072
Reconnaissance	1,695,132
Benign	51,989
Data theft	1,615

Decision: No change; documented.

Why before the merge: It describes what this dataset can teach.

Other datasets: See M12.

B6. DoS and DDoS flows are identical

Check: 5.4 behaviour duplicates | **Size:** 272,961 rows in 251 groups

The same 1-packet, 140-byte SYN to port 80 appears 122,631 times as DoS and 86,005 times as DDoS. DoS and DDoS differ in the **number of attacking machines**, which a single flow cannot show.

Both labels belong to the same family, *Denial of service*.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
7438830	6	80	1	140	0	0	0	2	0	DoS
65018	6	80	1	140	0	0	0	2	0	DDoS

Decision: Keep them; at family level there is no conflict.

Why before the merge: The conflict is between labels of this dataset only.

Other datasets: UNSW (U8), ToN-IoT (T7).

B7. Reconnaissance flows that look exactly like Benign flows

Check: 5.4 attack flows identical to Benign | **Size:** 378,185 Reconnaissance flows (22.3%); 57.6% of Benign is involved

Many scan probes are single packets answered by a reset, the same as a normal client reaching a closed port. In the example the same flow is Benign in one row and Reconnaissance in the other. As in ToN-IoT, the flow is too short to show the difference.

Example from the data

Row	PROTOCOL	L4_DST_PORT	IN_PKTS	IN_BYTES	OUT_PKTS	OUT_BYTES	FLOW_DURATION_MILLISECONDS	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	Attack
15352069	6	1	1	40	1	40	0	0	20	Benign
15211062	6	1	1	40	1	40	0	0	20	Reconnaissance

Decision: Keep them; do not apply the CIC-2018 rule (same reasoning as ToN-IoT T6).

Why before the merge: The decision depends on why the conflict exists in this dataset.

Other datasets: ToN-IoT (T6); CIC-2018 removes its own because the labels are wrong (C7).

5 After the merge: problems solved on the merged table

Each problem shows its size in the four datasets. The order of the columns is always UNSW, CIC-2018, ToN-IoT, BoT-IoT. Examples are taken from a random sample of 60,000 flows per dataset (`random_state=42`).

5.1 Decisions that change the data

M1. Codes stored as numbers

Check: 2.3 code columns (all four notebooks) | **Size:** 11 columns in every dataset

Ports, protocol numbers, TCP flags, ICMP and DNS types and FTP codes are identifiers stored as integers. Port 443 is not "more" than port 80, and an average protocol number has no meaning, but a model would treat them as quantities.

Example from the data

Row	dataset	L4_DST_PORT	PROTOCOL	CLIENT_TCP_FLAGS	DNS_QUERY_TYPE	Attack
1742981	UNSW	60365	6	27	0	Benign
19621116	CIC-2018	46388	6	20	0	Benign
18432811	ToN-IoT	53556	6	24	0	Benign
2783670	BoT-IoT	80	6	2	0	DDoS

Decision: Mark them as categorical on the merged table. The encoding (e.g. port ranges 0–1023 / 1024–49151 / 49152–65535) is chosen in feature engineering.

Why after the merge: The same rule, on the same columns, in every dataset: it is applied once, after the merge.

M2. Timestamps stored as integers, and they reveal the attack

Check: 2.2 + 7.3 time vs label | **Size:** 2 columns in every dataset

The start and end times are milliseconds since 1970. In every dataset the attacks were run at planned times: each CIC-2018 attack on its own day, ToN-IoT XSS only on 27 April 2019, BoT-IoT 90% of the flows on 4 June 2018. A model given the time would learn *when* the attack happened.

The four captures are also from different years (2015, 2018, 2018, 2019), so time alone would tell the dataset.

Example from the data

Dataset	capture period	hours with > 90% attacks
UNSW	22 Jan – 18 Feb 2015	–
CIC-2018	14 Feb – 2 Mar 2018	most attack days
ToN-IoT	23 – 29 Apr 2019	14 of 108
BoT-IoT	15 May – 19 Jun 2018	18 of 31

Decision: Convert to datetime on the merged table for the analysis (plots over time, a time-based split). **Never a model feature.**

M3. IP addresses reveal the label

Check: 7.2 IP vs label | **Size:** 2 columns in every dataset

In UNSW 4 machines send every attack; in CIC-2018 195 source IPs send only attacks. In ToN-IoT and BoT-IoT the attacker machines also send normal traffic, but 4 BoT-IoT machines still send 99.8% of the attacks.

An address says **who** sent a flow, not what the flow does, and on a new network (SQ3) the attackers have different addresses.

Example from the data

Dataset	source IPs	attack-only source IPs
UNSW	40	4
CIC-2018	181,876	195
ToN-IoT	15,270	2
BoT-IoT	20	0

Decision: Drop `IPV4_SRC_ADDR` and `IPV4_DST_ADDR` (the shared `clean()` already does it for each dataset; the merged table never contains them). No rows are removed.

Why after the merge: The same rule for every dataset.

M4. TTL is a fingerprint of the machines, not of the attack

Check: 7.5 TTL vs label + 3.6 TTL = 0 | **Size:** `MIN_TTL`, `MAX_TTL` in every dataset

The TTL depends on the operating system of the sender and on the number of routers: Linux starts at 64, Windows at 128. In UNSW benign traffic has 31/32 and attacks 63/254/255; in BoT-IoT almost every attack has 64. TTL identifies the attacker's computer.

The share of flows without a TTL (0 = not recorded) is also completely different per dataset, so TTL also reveals the dataset.

Example from the data

Dataset	% of flows with TTL = 0	typical attack TTL	typical Benign TTL
UNSW	1.15	63 / 254 / 255	31 / 32
CIC-2018	39.53	127 / 63	128 / 100–103
ToN-IoT	67.26	64	0 / 64 / 128
BoT-IoT	11.74	64	0 / 64 / 127

Decision: Keep the columns for the analysis; **not model features**. No rows are removed.

M5. Random identifiers

Check: 7.1 + 2.3 | **Size:** `L4_SRC_PORT`, `DNS_QUERY_ID` in every dataset

The source port is a free number chosen by the client's operating system, and the DNS query ID is a random 16-bit number. Neither describes behaviour; a model could only learn the habits of one attack tool.

Example from the data

Row	dataset	L4_SRC_PORT	L4_DST_PORT	DNS_QUERY_ID	Attack
409274	UNSW	55172	53	29217	Benign
2812862	CIC-2018	53149	53	46326	Benign
14069944	ToN-IoT	54997	53	17482	Benign
16632929	BoT-IoT	47171	53	3937	Benign

Decision: Not model features. Also left out when two flows are compared for identical behaviour.

M6. ICMP fields filled for TCP flows

Check: 3.6 disguised missing values | **Size:** 1.75 M / 1.54 M / 429 k / 9.8 k flows

`ICMP_TYPE` and `ICMP_IPV4_TYPE` only have a meaning for ICMP (protocol 1) and ICMPv6 (58), yet they hold values for many TCP flows. The values are leftovers of the exporter, not information about the flow.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
TCP flows with an ICMP value	1,754,341	1,538,930	429,291	9,776

Row	dataset	PROTOCOL	L4_DST_PORT	ICMP_TYPE	ICMP_IPV4_TYPE	Attack
1742981	UNSW	6	60365	21248	83	Benign
10206399	CIC-2018	6	80	41984	164	Benign
25260225	ToN-IoT	6	8080	12288	48	Backdoor
7405561	BoT-IoT	6	80	10240	40	DoS

Decision: Set both to 0 when `PROTOCOL` is neither 1 nor 58, on the merged table.

M7. `MIN_IP_PKT_LEN` = 0, and both `IP_PKT_LEN` columns copy other columns

Check: 4.2 + 4.3 rules | **Size:** 66 k / 1.98 M / 11.8 M / 12.1 M flows

No IP packet can be 0 bytes long, yet `MIN_IP_PKT_LEN` is 0 on one-way flows (`OUT_PKTS` = 0); `SHORTEST_FLOW_PKT` gives the correct value. Otherwise the two columns are identical, and `MAX_IP_PKT_LEN` equals `LONGEST_FLOW_PKT` on 100% of the flows of all four datasets.

A few one-way flows are exceptions with a non-zero `MIN_IP_PKT_LEN` (12 in CIC-2018, 60 in ToN-IoT, none in UNSW and BoT-IoT; found in Cyrine Chabbar's ToN-IoT check). They do not change the decision.

The wrong 0 is most common in floods and scans, so a model could learn the exporter's bug as a sign of attack.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows with MIN_IP_PKT_LEN = 0	66,121	1,975,355	11,838,122	12,102,947

Row	dataset	IN_PKTS	OUT_PKTS	SHORTEST_FLOW_PKT	MIN_IP_PKT_LEN	LONGEST_FLOW_PKT	MAX_IP_PKT_LEN	Attack
679421	UNSW	2	0	57	0	57	57	Generic
19621116	CIC-2018	1	0	40	0	40	40	Benign
18432811	ToN-IoT	1	0	329	0	329	329	Benign
13429529	BoT-IoT	9	0	28	0	28	28	DoS

Decision: Drop `MIN_IP_PKT_LEN` and `MAX_IP_PKT_LEN`; keep `SHORTEST_FLOW_PKT` and `LONGEST_FLOW_PKT`. Nothing is lost.

M8. Exact copies of other columns

Check: 7.4 redundancy | **Size:** 2 columns in every dataset

`TCP_FLAGS` = `CLIENT_TCP_FLAGS` OR `SERVER_TCP_FLAGS` and `ICMP_IPV4_TYPE` = `ICMP_TYPE` ÷ 256 on 100% of the flows of all four datasets.

Example from the data

Row	dataset	CLIENT_TCP_FLAGS	SERVER_TCP_FLAGS	TCP_FLAGS	Attack
1742981	UNSW	27	27	27	Benign
19621116	CIC-2018	20	0	20	Benign
18432811	ToN-IoT	24	0	24	Benign
2783670	BoT-IoT	2	20	22	DDoS

Decision: Drop `TCP_FLAGS` and `ICMP_IPV4_TYPE`; they can be recomputed at any time.

M9. Invalid DNS TTL values

Check: 4.2 rules + 6.3 limits | **Size:** UNSW 12, BoT-IoT 320 flows

RFC 2181 limits a DNS TTL to 2^{31}-1 seconds (68 years) and says larger values must be read as 0. Some values are close to 2^{32} (136 years): a negative number written as unsigned.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows above 2,147,483,647	12	0	0	320

Row	dataset	L4_DST_PORT	DNS_QUERY_TYPE	DNS_TTL_ANSWER	Attack
316940	UNSW	53	12	3,400,622,437	Exploits

Decision: Set values above 2^{31}-1 to 0 on the merged table, as the standard prescribes. Applied in the merge: 14 rows changed (UNSW 12; BoT-IoT 2, because its 320 flows had already been collapsed into 2 rows). One row then became identical to another row of the same dataset and label and was merged into it, its `count` added, so there is still one row per behaviour.

M10. Columns that reveal which dataset a flow comes from

Check: cross-dataset check (60,000-flow sample per dataset) | **Size:** best single column: MIN_TTL (78% correct, chance 25%)

After the merge a model could learn "this looks like ToN-IoT" instead of "this is an attack": the datasets have very different attack shares, so knowing the dataset already helps. We trained a tiny decision tree on **one column at a time** to predict the dataset. A column that does this well is a dataset fingerprint.

Some columns are fingerprints for a technical reason, not because of the traffic: the share of TTL = 0, of `FTP_COMMAND_RET_CODE` filled, of congestion flags or of non-zero `IAT_MIN` differs between the exporter set-ups.

Example from the data

Column	accuracy predicting the dataset
MIN_TTL	78.5
MAX_TTL	78.5
LONGEST_FLOW_PKT	74.1
TCP_WIN_MAX_IN	68.3
DURATION_IN	66.1
IN_BYTES	65.9
FLOW_DURATION_MILLISECONDS	65.7
SRC_TO_DST_IAT_MAX	64.7

Dataset	TTL = 0 %	FTP code non-zero %	flags > 31 %	IAT_MIN non-zero %	L7_PROTO decimal %	`MIN_IP_PKT_LEN = 0` %	duration 0 %
UNSW	1.15	18.58	0	0.08	1.01	2.81	5.26
CIC-2018	39.53	0	28.28	25.48	0	9.97	0
ToN-IoT	67.26	0.34	0.02	9.21	0	43.09	0
BoT-IoT	11.74	0	0	87.65	0.02	71.28	6.21

Decision: Exclude the technical fingerprints from the model features (TTL, already excluded by M4; `FTP_COMMAND_RET_CODE`). For the others, feature selection keeps a column only if it separates attacks **within** each dataset, not just the datasets. The model is always tested on a held-out dataset (SQ3).

M11. The same behaviour has a different family in another dataset

Check: cross-dataset behaviour comparison + merge (step 6) | **Size:** 17,587 behaviours; 38,453 rows of the merged table (none removed)

Each dataset alone is consistent for these behaviours, but the merged data is not: the same flow is, for example, Benign in one network and Reconnaissance in another. The largest group: about 14,200 behaviours that are Reconnaissance in BoT-IoT and Benign in ToN-IoT, two IoT testbeds where a short probe and a normal short connection look the same.

In the merge, a behaviour is flagged when it appears in at least two datasets with at least two different families: 17,587 behaviours, 38,453 rows. The rows are few, but the ToN-IoT ones stand for 5.3 million flows (common short Benign connections). UNSW has none: its behaviours do not occur in the other networks (see K13).

Removing them would delete whole groups of normal traffic from one dataset and attacks from another; keeping them unchanged teaches the model contradictions.

Example from the data

datasets : family (phase 2 check)	behaviours
BoT:Reconnaissance | ToN:Benign	14,234
BoT:Reconnaissance | CIC18:Exploitation and intrusion | ToN:Benign	258
CIC18:Benign | ToN:Web and injection	160
CIC18:Benign | ToN:Denial of service	72
CIC18:Exploitation and intrusion | ToN:Benign	40
CIC18:Benign | ToN:Reconnaissance	25

Dataset (merged table)	rows flagged	flows they stand for
UNSW	0	0
CIC-2018	2,203	133,484
ToN-IoT	20,554	5,346,962
BoT-IoT	15,696	83,986

Decision: Flag them with a column `ambiguous_across_sources` on the merged table (no row removed). Train with and without them and report both; the train/test split keeps each behaviour on one side only.

M12. One dataset dominates some families

Check: family × dataset, different behaviours kept (section 6) | **Size:** e.g. Web and injection: ToN-IoT 1,831,103 vs CIC-2018 53

Even with every behaviour kept, a family can come almost entirely from one network. A model would then learn that family as it looks in that network, and the comparison between networks (SQ2) would rest on one dataset.

Example from the data

Family	UNSW	CIC-2018	ToN-IoT	BoT-IoT
Benign	1,307,351	8,178,232	2,282,642	21,951
Credential attacks	0	94,890	697,124	0
Data theft	0	0	0	448
Denial of service	5,360	642,600	573,277	642,187
Exploitation and intrusion	42,740	27,213	0	0
Fuzzing and crypto	32,713	0	0	0
Interception and spoofing	0	0	4,951	0
Malware and botnets	3,802	2,804	23,471	0
Reconnaissance	13,134	0	225,074	898,729
Web and injection	0	53	1,831,103	0

Decision: Split train/test stratified by **dataset and family**, give each network the same total weight in training (section 6), and report results per dataset as well as overall. A family with fewer than 1,000 different behaviours in a dataset is not used to compare networks.

5.2 Decisions to keep it as it is

K1. TTL = 0 means "not recorded"

Check: 3.6 disguised missing values | **Size:** 1.1% / 39.5% / 67.2% / 11.6% of the flows

A packet with TTL 0 is thrown away by the network, so 0 means the exporter did not record it. It cannot be imputed: TTL depends on the sending machine.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows with TTL = 0	27,147	7,938,061	18,492,406	1,965,382

Decision: Keep 0 as the "unknown" marker. TTL is not a model feature anyway (M4).

K2. 0 means "does not apply" in protocol-specific fields

Check: 3.6 disguised missing values | **Size:** DNS and FTP fields

DNS fields are 0 outside DNS flows and FTP codes are 0 outside FTP. That is correct: there is nothing to record. Some DNS / FTP flows also have 0 because no answer came back.

Example from the data

Row	dataset	L4_DST_PORT	DNS_QUERY_TYPE	DNS_TTL_ANSWER	FTP_COMMAND_RET_CODE	Attack
409274	UNSW	53	1	60	0	Benign
2812862	CIC-2018	53	1	60	0	Benign
14069944	ToN-IoT	53	1	0	0	Benign
16632929	BoT-IoT	53	1	586	0	Benign

Decision: Keep 0 as "not applicable / no reply". No imputation.

K3. Byte counts too small for the number of packets

Check: 4.2 rules + 4.3 | **Size:** 45,540 / 0 / 53,198 / 80 flows

Every IP packet carries at least 20 bytes of header, so a 1-packet TCP flow cannot be 9 bytes. `SHORTEST_FLOW_PKT` contradicts `IN_BYTES` on these rows. The true value cannot be recovered.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows	45,540	0	53,198	80

Row	dataset	PROTOCOL	IN_PKTS	IN_BYTES	SHORTEST_FLOW_PKT	Attack
981040	UNSW	6	1	9	52	Benign
20335181	ToN-IoT	6	1	4	52	password

Decision: Keep the rows; document as an exporter error; re-check in the EDA.

K4. Size bins do not add up to the packet count

Check: 4.2 rules + 4.3 | **Size:** 482,792 / 1,100,188 / 148,670 / 1,356 flows

The five `NUM_PKTS_*` columns should add up to `IN_PKTS + OUT_PKTS`. In UNSW they count exactly 11 or 9 packets too many, a systematic exporter bug; in the other datasets the difference varies. Flows with an oversized packet (K6) count fewer, because that packet fits in no bin.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows	482,792	1,100,188	148,670	1,356

Row	dataset	IN_PKTS	OUT_PKTS	NUM_PKTS_UP_TO_128_BYTES	NUM_PKTS_128_TO_256_BYTES	Attack
188742	UNSW	11	11	31	0	Exploits
10966355	CIC-2018	1	1	2	2	Benign
20282847	ToN-IoT	3	3	11	0	password
15281736	BoT-IoT	6	6	9	1	Reconnaissance

Decision: Keep; document. In feature engineering, use the bins as shares of their own total.

K5. Largest gap between packets longer than the flow

Check: 4.2 rules | **Size:** 33,970 / 547,302 / 106,238 / 111 flows

`SRC_TO_DST_IAT_MAX` (the longest pause between two packets) can never be longer than the whole flow. On these flows it is, sometimes by seconds (a 1-ms flow with a 3,333-ms gap): the exporter measures the two values with different clocks or cuts the flow differently. The true values cannot be recovered.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows	33,970	547,302	106,238	111

Row	dataset	IN_PKTS	FLOW_DURATION_MILLISECONDS	SRC_TO_DST_IAT_MAX	Attack
2339928	UNSW	9	1,210	1,652	Benign
10966355	CIC-2018	1	1	147	Benign
20335181	ToN-IoT	1	1	3,333	password

Decision: Keep; document.

K6. Packets above the Ethernet limit of 1514 bytes

Check: 4.2 rules + 4.3 | **Size:** 0 / 1,002 / 554 / 1,075 flows

A normal Ethernet frame carries at most 1514 bytes. Larger "packets" (up to 65,212 bytes) appear when the network card merges segments before the capture tool sees them. A capture artefact, not attack behaviour.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows	0	1,002	554	1,075

Row	dataset	PROTOCOL	IN_PKTS	LONGEST_FLOW_PKT	Attack
15009988	CIC-2018	6	5	1,913	Infilteration
15281736	BoT-IoT	6	6	2,948	Reconnaissance

Decision: Keep; document.

K7. Small rule violations

Check: 2.5 + 4.2 | **Size:** a few flows each

Rare cases that are allowed by the protocols or too few to matter: TCP flows without any flag (a NULL scan sends exactly this), port 0, a few invalid DNS / ICMP type values in Benign traffic, `DURATION_IN`/`DURATION_OUT` 1 ms longer than the flow (rounding), and one-way single-packet flows whose packet is longer than `IN_BYTES` (found in Cyrine Chabbar's ToN-IoT check).

Example from the data

Rule	UNSW	CIC-2018	ToN-IoT	BoT-IoT
TCP flow without flags	7	231	1	5,061
TCP/UDP port 0	6	18	4,782	1,171
invalid DNS / ICMP type	12	0	18	0
duration 1 ms too long	0	4,106	17	7,154
1-packet flow longer than IN_BYTES	0	11	17	0

Decision: Keep. BoT-IoT's 3,872 Benign TCP flows without flags (7.4% of its Benign) are re-checked in the EDA.

K8. Unusual but legitimate protocol values

Check: 2.3 + 3.6 | **Size:** ICMPv6, LLMNR, congestion flags

ICMPv6 (protocol 58): normal IPv6 neighbour discovery, so ICMP fields are valid there. **LLMNR** (port 5355): Windows name lookups in the DNS format, so a DNS query type is valid there. **ECE/CWR** flags (TCP flags above 31): valid congestion flags.

They are distributed very unevenly between datasets (28% of CIC-2018 flows have ECE/CWR, almost none elsewhere), which is why they are also checked in M10.

Example from the data

Value	UNSW	CIC-2018	ToN-IoT	BoT-IoT
ICMPv6 flows	0	18,745	1,255	94
LLMNR flows	0	102,553	13,055	26
flows with ECE/CWR	0	28%	7,290	836

Decision: Keep them.

K9. No unique flow identifier

Check: 5.3 flow key | **Size:** 483,517 / 1,344 / 1,306,753 / 612 rows share a key

NetFlow has no flow ID. The natural key (addresses, ports, protocol, start time) is shared by rows that are different flows, because the timestamps only have millisecond resolution and the exporter splits long conversations.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
non-duplicate rows sharing a key	483,517	1,344	1,306,753	612

Decision: No action; never deduplicate on the key. Only exact duplicates are removed (U4, C4, T4).

K10. Many statistical outliers

Check: 6.1 IQR / z-score + 6.2 box plots | **Size:** IQR flags 5–25% of the flows per column

Traffic is extremely skewed (skewness up to 4,953): most flows are tiny and a few are huge. The IQR rule flags whatever is far from the majority class, so in UNSW 100% of attacks are "outliers" in TTL, and in BoT-IoT (99.7% attacks) Benign is the outlier.

No physical limit is broken (6.3) apart from K6 and M9: the extremes are real traffic and often the attack itself.

Decision: Keep every outlier; use a log transform in feature engineering.

K11. Near-constant columns differ per dataset

Check: 7.1 near-constant | **Size:** e.g. `SRC_TO_DST_IAT_MIN`, `FTP_COMMAND_RET_CODE`

A column that is constant in one dataset can be informative in another: `SRC_TO_DST_IAT_MIN` is 99.9% zero in UNSW but has a median of 3,115 ms in BoT-IoT; the FTP code is 100% zero in CIC-2018 and BoT-IoT but real in UNSW and ToN-IoT; BoT-IoT's retransmission columns are 99.94% zero.

Example from the data

Column	UNSW	CIC-2018	ToN-IoT	BoT-IoT
SRC_TO_DST_IAT_MIN	99.9% zero	not constant	not constant	not constant
FTP_COMMAND_RET_CODE	real codes	100% zero	99.7% zero	100% zero

Decision: Keep for now; decide in feature selection on the merged data (the FTP code is excluded by M10).

K12. Very high correlations between columns

Check: 7.4 correlation | **Size:** pairs above 0.98, different in each dataset

Some columns move together almost perfectly (e.g. `FLOW_DURATION_MILLISECONDS` and `DURATION_IN` in all four datasets; `OUT_BYTES` and `OUT_PKTS` in BoT-IoT). Correlated is not identical, and the pairs differ per dataset.

Decision: Decide in feature selection on the merged data, with one rule for the whole team.

K13. Normal traffic looks different in every network

Check: cross-dataset Benign comparison | **Size:** medians and overlap of Benign behaviours

A typical Benign flow is a different thing in each network: an office network, an IoT testbed and a lab have different applications and devices. Almost no Benign behaviour of one network appears in another; the only real overlap is ToN-IoT with CIC-2018 (28%). Second table: % of the row's Benign flows whose exact behaviour also appears as Benign in the column.

This is not an error to fix; it is exactly what makes detection on an unseen network hard.

Example from the data

Dataset	IN_BYTES	IN_PKTS	FLOW_DURATION_MILLISECONDS	LONGEST_FLOW_PKT	**flows**
UNSW	1,684	19	28	510	2,237,731
CIC-2018	128	2	39	157	17,514,626
ToN-IoT	52	1	1	52	16,792,214
BoT-IoT	40	1	0	40	51,989

Benign of	**UNSW**	**CIC-2018**	**ToN-IoT**	**BoT-IoT**
UNSW	100	0	0	0
CIC-2018	0	100	1.65	0.07
ToN-IoT	0	28.45	100	0.02
BoT-IoT	0	0.18	1.62	100

Decision: Keep. It is measured, not corrected: SQ3 tests a model on a dataset it has not seen, and K13 is the explanation if it fails.

K14. The same attack family behaves differently between networks

Check: cross-dataset DoS comparison | **Size:** Denial of service in all four datasets

Denial of service is the only family present in all four datasets, but the tools and therefore the flows differ: a typical ToN-IoT DDoS flow is 2 packets lasting 1 ms, a BoT-IoT DoS flow is 6 packets lasting 31 s, a UNSW DoS flow 10 packets in half a second, and CIC-2018 ranges from 1-packet Slowloris flows to 7 MB LOIC-UDP floods lasting two minutes. The medians below differ by orders of magnitude.

Example from the data

Dataset: label	IN_BYTES	IN_PKTS	FLOW_DURATION_MILLISECONDS	LONGEST_FLOW_PKT	flows
UNSW: DoS	792	10	561	368.5	5,980
CIC-2018: DDoS HOIC	521	5	20	975	1,032,311
CIC-2018: DDoS LOIC-UDP	7,226,700	120,445	120,551	60	3,450
CIC-2018: DDoS LOIC-HTTP	232	5	20,577	1,004	288,589
CIC-2018: DoS GoldenEye	689	5	7,007	1,024	61,300
CIC-2018: DoS Hulk	13,784	109	85,996	987	100,076
CIC-2018: DoS SlowHTTPTest	60	1	1	60	105,550
CIC-2018: DoS Slowloris	282	1	1	282	36,040
ToN-IoT: DDoS	112	2	1	60	4,141,256
ToN-IoT: DoS	3,720	93	150	40	203,456
BoT-IoT: DDoS	420	6	20,067	140	7,150,882
BoT-IoT: DoS	360	6	31,030	28	8,034,190

Family: dataset label	IN_PKTS	IN_BYTES	OUT_BYTES	FLOW_DURATION_MILLISECONDS
Credential: CIC-2018 SSH brute force	23	3,148	3,869	375
Credential: CIC-2018 FTP brute force	1	60	40	1
Credential: ToN-IoT Password	5	415	921	20
Web: CIC-2018 XSS	6	3,168	0	89,479
Web: CIC-2018 SQL injection	6	3,168	0	89,479
Web: ToN-IoT XSS	5	574	334	26
Web: ToN-IoT Injection	5	514	2,239	289
Malware: CIC-2018 Bot	5	538	341	11
Malware: ToN-IoT Backdoor	6	288	240	1,500
Recon: ToN-IoT Scanning	1	44	0	1
Recon: BoT-IoT Reconnaissance	1	44	40	3

Decision: Keep. This is the question SQ2 asks; the EDA compares the families across networks with statistical tests. The second table extends the comparison to the other shared families (office network CIC-2018 against the IoT networks): Credential attacks and Web attacks also differ in service and size, while Reconnaissance looks alike in the two IoT networks and bot / backdoor traffic looks alike across both kinds of network (port 8080, 5–6 packets).

K15. Two target columns

Check: 1.2 `Label` vs `Attack` | **Size:** 0 disagreements in all four datasets

`Label` (0/1) agrees with `Attack` on every row of every dataset. `map_labels()` adds `is_attack`, which is identical to `Label`; the targets after cleaning are `is_attack`, `attack_family` and `attack_label`.

Decision: Drop `Label` after `map_labels()` (two copies of the same target only invite mistakes); keep the raw `Attack` name for traceability. None of these columns is ever a model feature.

K16. Errors in the official documentation

Check: 0.3 feature dictionary | **Size:** one document for all four datasets

`NetFlow_v3_Features.csv` has three IAT descriptions copied from the minimum ("Minimum" for MAX, AVG and STDDEV), `DURATION_OUT` described as client to server, `..._SECOND_BYTES` described as bytes per second (it is per millisecond, direction swapped), trailing spaces in the names, typos ("Arrval", "Sandard Deviaion"), and broken characters in the description of `NUM_PKTS_1024_TO_1514_BYTES` ("> ?? 1024", an encoding error, found by Cyrine Chabbar). The BoT-IoT web page also has wrong row counts (B4).

Example from the data

Column	official description	correct meaning
DST_TO_SRC_IAT_MAX	Minimum Inter-Packet Arrval Time (dst > src)	Maximum gap between packets, server to client
DURATION_OUT	Client to Server stream duration (msec)	Server to client stream duration
SRC_TO_DST_SECOND_BYTES	Src to dst Bytes/sec	OUT bytes per millisecond

Decision: Report them; use our corrected descriptions.

K17. Average gap between packets smaller than the minimum gap

Check: 4.4 extra rules (found in Cyrine Chabbar's ToN-IoT check) | **Size:** 316 / 862,525 / 456,179 / 43,806 flows

An average can never be smaller than the minimum of the same values, yet `..._IAT_AVG` < `..._IAT_MIN` on these flows. The arithmetic shows why: a CIC-2018 flow with 4 packets has minimum = maximum = 70 ms but an average of 23 ms over a 71 ms flow. Three gaps that add up to 70 ms and average 23 ms are 0, 0 and 70 ms: the packets came in a burst that rounds to 0 ms.

So the **average is right** and the **minimum skips the 0-ms gaps** (it reports the smallest non-zero gap). It is an exporter convention, not corrupted data; it is most common in bursty traffic (DDoS, XSS, HOIC).

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
flows	316	862,525	456,179	43,806

Row	IN_PKTS	SRC_TO_DST_IAT_MIN	SRC_TO_DST_IAT_AVG	OUT_PKTS	DST_TO_SRC_IAT_MIN	DST_TO_SRC_IAT_AVG	Attack	dataset
9656	1	0	0	3	6	1	Benign	UNSW
47	5	6	1	5	0	2	Benign	CIC-2018
6436	2	718	718	3	718	359	Benign	ToN-IoT
8	6	3,738	3,557	4	3,738	2,913	DDoS	BoT-IoT

Decision: Keep the rows and values. Read `IAT_MIN` as "smallest non-zero gap" in the report; feature selection decides whether `IAT_MIN` is kept (K11).

K18. TCP window 0 although packets were sent

Check: 4.4 extra rules (found in Cyrine Chabbar's ToN-IoT check) | **Size:** 45,884 / 1,371,482 / 3,082,721 / 2,936,334 flows

The TCP window is the amount of data a side is ready to receive; 0 looks like "not recorded", as TTL = 0 does.

For CIC-2018, ToN-IoT and BoT-IoT the cause is legitimate: in the 60,000-flow samples, 100% of the server-side cases (89–100% of the client-side ones) are flows where that side sent a **reset** (RST), which carries window 0 by design. These are refused connections: the SYN floods and scans of BoT-IoT (2.9 million) and ToN-IoT. UNSW is the exception: its 45,833 client-side zeros contain no reset, so there the 0 is unexplained.

Example from the data

	UNSW	CIC-2018	ToN-IoT	BoT-IoT
TCP flows with window 0 despite packets	45,884	1,371,482	3,082,721	2,936,334
client side (IN)	45,833	751,664	545,492	6,042
server side (OUT)	51	620,974	2,546,494	2,930,292

Row	dataset	OUT_PKTS	TCP_WIN_MAX_OUT	SERVER_TCP_FLAGS	Attack
1514968	UNSW	1	0	24	Benign
711896	CIC-2018	1	0	20	FTP-BruteForce
8055771	ToN-IoT	1	0	20	Benign
2783670	BoT-IoT	1	0	20	DDoS

Decision: Keep. A window of 0 on a reset is real behaviour (and a useful sign of refused connections). In UNSW, treat the client-side 0 as "unknown" in the EDA.

6 Sampling, split and evaluation plan

This section turns the decisions of sections 1–5 into one plan for phase 3. It replaces the earlier rule "at most 200,000 flows per label": that cap would have kept only 15–40% of the different attack behaviours of the large classes and turned Benign into 15% of the data. The plan below was chosen without a cost limit; it still fits in memory (about 4 GB).

6.1 The rule

1. **Clean each dataset first** (sections 1–4): mislabelled rows removed (C5, C6, C7, T5), broken columns dropped.
2. **One row per different behaviour, Benign and attack alike, no upper limit.** Only exact repeats are removed (a behaviour is everything the flow did, leaving out the IP addresses, source port, timestamps and DNS query ID). A `count` column keeps how many times each behaviour appeared. It is **never** a model feature. If one behaviour has two labels inside a dataset, it keeps one row per label.
3. **Balance with weights, not by deleting rows:** class weights so that every class counts equally (rare attacks such as Worms count as much as Benign), and network weights so that every dataset counts equally (CIC-2018's 8 million Benign rows do not outweigh UNSW).
4. **Grouped split 80 / 20 by behaviour:** a behaviour is never in both the training and the test set.
5. **Realistic test sets: 95% Benign / 5% attacks,** the way a real network looks; also reported at 99 / 1 to show how false alarms grow when attacks are rarer.

6.2 The numbers

Raw data against the final dataset (one row per behaviour, after the before-merge removals). The data shrinks from 67 to about 17.6 million rows, and no different behaviour is lost; Benign becomes the majority again, as in most real networks.

Dataset	raw Benign	raw attack	raw Benign %	final Benign	final attack	final total	final Benign %
UNSW	2,237,731	127,693	94.6%	1,307,351	97,749	1,405,100	93.0%
CIC-2018	17,514,626	2,600,903	87.1%	8,178,232	767,560	8,945,792	91.4%
ToN-IoT	16,792,214	10,728,046	61.0%	2,282,642	3,355,000	5,637,642	40.5%
BoT-IoT	51,989	16,881,819	0.3%	21,951	1,541,364	1,563,315	1.4%
All four	36,596,560	30,338,461	54.7%	11,790,176	5,761,673	17,551,849	67.2%

The cleaning produced exactly these numbers. The merge then merged one more row (decision M9), so the merged table has **17,551,848 rows** standing for 63,012,224 flows (the raw flows minus the exact duplicates and the mislabelled flows removed in sections 2 and 3).

6.3 Is it enough?

What a model learns from is the number of **different behaviours**, not the number of flows: FTP-BruteForce has 386,720 flows but one single behaviour. Thresholds: at least 1,000 different behaviours = enough to learn and report the label; 100–999 = small (label with a higher class weight, always also reported at family level); under 100 = too small (scored only inside its family). Result: 29 attack labels are enough, 3 small, 2 too small, and 2 removed entirely by the before-merge decisions:

Label	flows	different behaviours kept	status	family
UNSW: Worms	158	146	small	Malware and botnets
CIC-2018: FTP brute force	386,720	1	too small	Credential attacks
CIC-2018: DoS SlowHTTPTest	105,550	0	removed (C5-C7)	Denial of service
CIC-2018: Web brute force	1,618	910	small	Credential attacks
CIC-2018: XSS	480	0	removed (C5-C7)	Web and injection
CIC-2018: SQL injection	440	53	too small	Web and injection
BoT-IoT: Theft	1,615	448	small	Data theft

Every family has enough data except *Data theft* (Theft, BoT-IoT only): it can be learned, but it cannot be tested on an unseen network because no other dataset contains it. Small labels are reported with repeated grouped cross-validation (5 folds), because one test split holds only a few dozen of their behaviours. Synthetic data (e.g. SMOTE) is never used in a test set.

6.4 Why the repeat count is not used as the training weight

Training with `weight = count` is mathematically the same as keeping all the copies: a row with weight 41,000 counts like 41,000 identical rows, so the floods would dominate again. The count is used in two other ways: optionally damped as `log(1 + count)` in training, and as the weight in a second evaluation that measures the model on the traffic as it really occurred. This follows common practice: removing duplicates is how NSL-KDD repaired KDD Cup 99 (Tavallaee et al., 2009), and ignoring the real attack rate is listed as a pitfall in Arp et al., "Dos and Don'ts of Machine Learning in Computer Security" (USENIX Security 2022).

6.5 Split, tests and metrics

- **Test 1, same network:** train and test inside one dataset (the best case).
- **Test 2, all networks mixed:** the merged data, split stratified by dataset and family.
- **Test 3, unseen network (SQ3):** train on three datasets, test on the fourth, four rounds. Everything learned from data (scaling, feature selection, the M10 fingerprint check) is decided on the three training datasets only.
- **Metrics:** macro-F1, recall per class, Benign recall and the false-alarm rate on the realistic test sets; each at three levels: attack or not, family, label. A family is only scored on a test network that contains it.

6.6 The order of phase 3

1. For each dataset: its own before-merge decisions (column fixes, exact duplicates, wrong labels) → one row per behaviour with `count` → `clean()` → `map_labels()`, then drop `Label` (identical to `is_attack`) → checks → save. The collapse runs **before** `clean()` drops the IP addresses, so two machines doing the same thing still count as two flows.
2. Because exact duplicates are removed first, C6 and C7 remove fewer rows in the cleaning than counted in phase 2 (CIC-2018: 30 instead of 60 port-21 SSH flows, 95,558 instead of 150,600 Benign-identical attack flows). These are the same flows minus their duplicates; the result is unchanged.
3. Stack the four files; add `network_type` from `DATASETS` in `config.py` ("Mixed lab", "Enterprise / office", "IoT / industrial", "IoT botnet") for the analysis; like `source_dataset`, it is **never** a model feature.
4. Apply the **fixed** after-merge rules (section 5.1: columns dropped, ICMP fields to 0, DNS TTL rule, `ambiguous_across_sources` flag) and run `validate()` again.
5. Split (grouped by behaviour; one dataset held out for test 3).
6. Fit everything **learned from data** on the training part only: scaling, log transform, feature selection, correlation threshold.

6.7 Limits

- BoT-IoT has almost no normal traffic (21,951 different Benign behaviours); its realistic test set is small and its Benign results are reported with that caveat.
- ToN-IoT is the largest part of the final data because its attacks are very varied; network weights and per-dataset results keep it from hiding the others.
- 95 / 5 is an assumption about real networks; the 99 / 1 results show how sensitive the conclusions are to it.
- A "different behaviour" means exactly equal in every column; two floods that differ by 1 ms count as two. "100% covered" is therefore a strict, safe statement.